

Continuous Hidden Markov Models with Poisson Jump Processes for Financial Time Series Generation

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April 11, 2026

Abstract

We present a continuous hidden Markov model (CHMM) framework for generating synthetic financial time series that preserve the stylized facts of asset returns. Building on the discrete-state approach of [Alswaidan and Varner \[2026\]](#), we learn Gaussian emission parameters and transition probabilities directly from continuous observations via the Baum-Welch (Expectation-Maximization) algorithm, eliminating the quantization error inherent in Laplace quantile binning while retaining the Poisson jump-duration mechanism for volatility clustering. Applied to ten years of daily SPDR S&P 500 ETF (SPY) data, the framework achieves KS pass rates above 87% across all state resolutions ($K \in \{3, 6, 9, 11, 13\}$) both in-sample and out-of-sample, with distributional fidelity improving monotonically with K . Baseline comparisons confirm that Gaussian i.i.d. generators fail entirely (0% KS), while the CHMM variants consistently outperform Laplace i.i.d. generators and match or exceed bootstrap resampling on distributional metrics. Cross-asset generalization to NVDA, JNJ, and JPM confirms that the framework adapts to diverse equity risk profiles.

We further extend the framework to *volatility indices* by applying it to CBOE VIX log returns—an asset class exhibiting distinct statistical properties including stronger mean reversion, positive skewness, and pronounced volatility clustering. The VIX analysis achieves KS pass rates above 99% at higher state resolutions ($K \geq 9$), with the jump mechanism becoming essential as K increases: the base CHMM-NJ model already captures volatility clustering effectively at small K , but at larger K the learned transition matrix produces insufficient tail-state residence times, necessitating the Poisson jump-duration extension. The continuous emission structure provides a natural foundation for downstream stochastic volatility applications; we outline how the learned regime-dependent emission variances can parameterize the mean-reversion target in a modified Heston model, enabling closed-form implied volatility generation for option pricing.

Keywords: Continuous Hidden Markov Model, Baum-Welch Algorithm, Jump-Diffusion, Volatility Clustering, Synthetic Financial Data, Regime-Switching, VIX, Implied Volatility

1 Introduction

Generating synthetic financial time series that faithfully preserve the statistical properties of real market data is a central challenge in quantitative finance. High-fidelity synthetic data are essential for stress testing risk models against plausible but unobserved market scenarios,

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validating portfolio optimization algorithms, and augmenting limited training sets for machine learning pipelines [Assefa et al., 2020, Jordon et al., 2022]. Empirical equity excess growth rates exhibit three well-documented statistical regularities—heavy-tailed (leptokurtic) distributions, negligible linear autocorrelation in raw returns, and persistent volatility clustering—that any generative framework must simultaneously reproduce to be considered statistically faithful [Cont, 2001, Mandelbrot, 1963].

Alswaidan and Varner [2026] addressed this challenge with a discrete hidden Markov model (HMM) that partitioned continuous excess growth rates into Laplace quantile-defined states and estimated transitions by direct frequency counting. A Poisson jump-duration mechanism forced the model to dwell in high-volatility tail states for empirically realistic durations, partially reproducing volatility clustering. That framework achieved high distributional pass rates and scaled to a 424-asset pipeline, but the discretization step introduced quantization error: continuous observations were binned into categorical states, and the emission model sampled from within-bin Student- t distributions rather than learning the emission parameters jointly with the transition structure.

In this study, we extend that framework to *continuous Gaussian emissions* learned via the Baum-Welch (Expectation-Maximization) algorithm [Baum et al., 1970, Rabiner and Juang, 1986]. Each hidden state emits from a Normal distribution $\mathcal{N}(\mu_k, \sigma_k)$ whose parameters are estimated iteratively alongside the transition matrix, eliminating the need for quantile binning and enabling the model to capture fine-grained regime dynamics directly from the data. The Poisson jump-duration mechanism is preserved, operating on the continuous model’s state space to enforce tail-state persistence.

The key contributions of this work are:

1. **Baum-Welch training** of emission means, variances, and transition probabilities directly from continuous observations, with a quantile-based initialization strategy that avoids degenerate local optima.
2. **State resolution analysis** across $K \in \{3, 6, 9, 11, 13\}$ hidden states, demonstrating that distributional fidelity improves monotonically with K while temporal structure (ACF-MAE) remains stable.
3. **Comprehensive validation** using five complementary metrics—Kolmogorov-Smirnov (KS) pass rates, excess kurtosis matching, autocorrelation mean absolute error (ACF-MAE), Wasserstein-1 distance, and Hellinger distance—on both in-sample (2014–2024) and out-of-sample (2025) data.
4. **Cross-asset generalization** to NVDA, JNJ, and JPM, confirming that the framework adapts to diverse equity risk profiles without modification.
5. **Extension to volatility indices**: application of the CHMM framework to CBOE VIX log returns, an asset class with distinct statistical properties (positive skewness, mean reversion, extreme kurtosis), demonstrating that the framework generalizes beyond equity returns to volatility dynamics. The VIX analysis further clarifies the role of the jump mechanism: at small K , the base model already captures volatility clustering, but as K increases the learned transition matrix produces insufficient tail-state persistence, making the Poisson jump extension essential.
6. **Pathway to implied volatility modeling**: the regime-dependent emission variances learned by the CHMM provide a natural parameterization for the mean-reversion target $\theta(S_t)$ in a modified Heston stochastic volatility model, enabling closed-form implied volatility generation for option pricing without circular dependence on market-observed IV surfaces.

The continuous emission structure provides a natural foundation for stochastic volatility extensions that the discrete quantile-binning approach cannot support. In particular, the VIX application demonstrates that the CHMM framework can directly model the volatility process itself—not merely generate synthetic returns—opening a pathway to regime-aware option pricing via integration with the Heston model [Heston, 1993].

2 Related Work

Research on reproducing the canonical stylized facts of financial markets spans parametric return models, latent-state frameworks, and deep generative approaches. This section reviews the key developments that motivate the present work.

2.1 Stylized Facts and Parametric Models

The Black-Scholes-Merton framework [Black and Scholes, 1973] assumes log-returns are i.i.d. Gaussian, but this conflicts with well-documented empirical regularities. Mandelbrot [1963] demonstrated that speculative price changes exhibit heavy tails (leptokurtosis), Fama [1970] confirmed negligible linear autocorrelation in raw returns, and Schwert [1989] showed that the autocorrelation of absolute returns decays slowly—the signature of volatility clustering. Cont [2001] provides a systematic review confirming these regularities across asset classes and sampling frequencies.

GARCH models [Engle, 1982, Bollerslev, 1986] capture conditional variance dynamics but require fat-tailed noise distributions to produce heavy tails and cannot represent discrete market regimes. Stochastic volatility models [Heston, 1993] treat volatility as a latent process but require computationally intensive estimation. Jump-diffusion models [Merton, 1976, Kou, 2002] add Poisson-driven jumps to the price process, producing heavy-tailed marginals, but lack a mechanism for volatility persistence: post-jump volatility returns immediately to its baseline level.

2.2 Hidden Markov Models and Regime-Switching

Hamilton [1989] introduced the regime-switching HMM to economics, showing that U.S. GNP growth is well described by a two-state model. HMMs have since become a standard tool for latent-state modeling of financial time series [Kim and Nelson, 1999, Nguyen, 2018, Rossi and Gallo, 2006]. Rydén et al. [1998] systematically evaluated HMMs against stylized facts: Gaussian-emission HMMs with few states reproduce heavy tails and the absence of return autocorrelation, but fail to capture volatility clustering because the ACF of absolute returns decays too quickly. Bulla and Bulla [2006] showed that hidden semi-Markov models (HSMMs) improve volatility-clustering reproduction by allowing flexible sojourn-time distributions.

The discrete HMM framework of Alswaidan and Varner [2026] addressed the volatility-clustering limitation by adding a Poisson jump-duration mechanism that forced the model to dwell in tail states for empirically realistic durations. That framework used Laplace quantile binning and direct frequency counting, bypassing the Baum-Welch algorithm entirely. The present work replaces the discretization step with continuous Gaussian emissions learned via Baum-Welch, preserving the jump mechanism while eliminating quantization error, and extends the framework to volatility indices (VIX) whose statistical properties—positive skewness, mean reversion, extreme kurtosis—differ markedly from equity returns.

2.3 The Baum-Welch Algorithm

The Baum-Welch algorithm [Baum et al., 1970] is the standard EM procedure for HMM parameter estimation. It alternates between an E-step (computing posterior state probabilities via

the forward-backward algorithm) and an M-step (re-estimating transition probabilities, emission parameters, and the initial state distribution). [Bilmes \[1998\]](#) provides a tutorial treatment. A well-known limitation is sensitivity to initialization: poor starting values can lead to degenerate local optima where all states collapse to the global mean [\[Rabiner and Juang, 1986\]](#). We address this with a quantile-based initialization strategy that seeds each state’s emission parameters from a distinct segment of the sorted observation sequence.

2.4 Deep Generative Models

Deep generative models have attracted interest for financial time series generation. [Takahashi et al. \[2019\]](#) found that GAN architectures produced sequences that visually resembled financial data but failed to reproduce ACF structure. [Kwon and Lee \[2024\]](#) confirmed that volatility clustering remained elusive unless the architecture was specifically designed for it. [Alswaidan and Varner \[2026\]](#) benchmarked a GRU-based generator against HMM variants and found that it closed 83% of the ACF-MAE gap but suffered variance collapse (0.6% KS pass rate). The present study focuses on the parametric HMM approach, which offers interpretability and computational efficiency advantages over neural generators.

3 Methods

3.1 Data

The empirical foundation of this study is a dataset of 424 United States-listed equities and exchange-traded funds spanning ten years (2014–2024). We focus our single-asset equity analysis on the SPDR S&P 500 ETF Trust (SPY), which tracks the S&P 500 index. The data include daily open, high, low, and close prices along with volume-weighted average price (VWAP) metrics. The in-sample dataset comprises $T = 2,766$ trading days (January 3, 2014 to December 31, 2024). A subsequent 219 trading days in 2025 are reserved for out-of-sample testing.

To extend the framework beyond equity returns, we additionally apply it to the CBOE Volatility Index (VIX), which measures 30-day implied volatility of S&P 500 options and is widely regarded as the market’s “fear gauge.” Daily VIX close prices spanning April 2006 to December 2024 ($T = 4,713$ trading days) constitute the in-sample dataset, with January 2025 to April 2026 ($T = 317$ days) reserved for out-of-sample testing. VIX log returns exhibit statistical properties that differ markedly from equity returns—positive skewness (reflecting asymmetric volatility spikes), pronounced mean reversion, and extreme excess kurtosis—making it a challenging test case that probes the generality of the CHMM framework across asset classes.

3.2 Excess Growth Rate Calculation

We compute the excess growth rate $G_{i,j}$ for ticker i between time periods $j - 1 \rightarrow j$ given the equity price series $P_{i,j}$:

$$G_{i,j} \equiv \frac{1}{\Delta t} \cdot \ln \left(\frac{P_{i,j}}{P_{i,j-1}} \right) - r_f \quad (1)$$

where $\Delta t = 1/252$ (one trading day in years) and r_f is a constant risk-free rate. The excess growth rate has units of year^{-1} and is time-additive under continuous compounding.

3.3 Continuous Hidden Markov Model

We define the hidden Markov model as the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$ [\[Rabiner and Juang, 1986\]](#), where:

- $S_t \in \mathcal{S} = \{1, \dots, K\}$ are the hidden states representing market regimes,

- $\mathcal{O} \subset \mathbb{R}$ is the continuous observation space of excess growth rates,
- $\mathbf{T} \in \mathbb{R}^{K \times K}$ is the state transition matrix with $T_{ij} = \mathbb{P}(S_{t+1} = j \mid S_t = i)$,
- $\mathbf{E} = \{(\mu_k, \sigma_k)\}_{k=1}^K$ specifies the Gaussian emission distributions $G_t \mid S_t = k \sim \mathcal{N}(\mu_k, \sigma_k^2)$,
- $\bar{\pi}$ is the stationary distribution satisfying $\bar{\pi} = \bar{\pi}\mathbf{T}$.

Unlike the discrete approach of [Alswaidan and Varner \[2026\]](#), which discretizes observations into Laplace quantile bins and estimates $\mathbf{T}$ by frequency counting, we learn $\mathbf{T}$, $\{\mu_k\}$, $\{\sigma_k\}$, and $\bar{\pi}$ jointly from the continuous observations via the Baum-Welch algorithm.

3.4 Baum-Welch (EM) Algorithm

The Baum-Welch algorithm [[Baum et al., 1970](#)] iterates between two steps until convergence.

E-Step. Compute the forward variables $\alpha_t(k) = \mathbb{P}(O_1, \dots, O_t, S_t = k \mid \mathcal{M})$ and backward variables $\beta_t(k) = \mathbb{P}(O_{t+1}, \dots, O_T \mid S_t = k, \mathcal{M})$ via:

$$\log \alpha_t(j) = \text{logsumexp}(\log \alpha_{t-1}(\cdot) + \log T_{\cdot,j}) + \log b_j(O_t) \quad (2)$$

$$\log \beta_t(i) = \text{logsumexp}(\log T_{i,\cdot} + \log b_{\cdot}(O_{t+1}) + \log \beta_{t+1}(\cdot)) \quad (3)$$

where $b_k(O_t) = \phi(O_t; \mu_k, \sigma_k)$ is the Gaussian emission density. All computations are performed in log-space to prevent numerical underflow for long sequences [[Bilmes, 1998](#)].

The posterior state probability is:

$$\gamma_t(k) = \mathbb{P}(S_t = k \mid O_{1:T}, \mathcal{M}) = \frac{\alpha_t(k)\beta_t(k)}{\sum_{j=1}^K \alpha_t(j)\beta_t(j)} \quad (4)$$

M-Step. Re-estimate parameters using the posterior probabilities:

$$\mu_k^{\text{new}} = \frac{\sum_{t=1}^T \gamma_t(k) \cdot O_t}{\sum_{t=1}^T \gamma_t(k)} \quad (5)$$

$$(\sigma_k^{\text{new}})^2 = \frac{\sum_{t=1}^T \gamma_t(k) \cdot (O_t - \mu_k^{\text{new}})^2}{\sum_{t=1}^T \gamma_t(k)} \quad (6)$$

$$T_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)} \quad (7)$$

where $\xi_t(i, j) = \mathbb{P}(S_t = i, S_{t+1} = j \mid O_{1:T}, \mathcal{M})$.

Initialization. EM is sensitive to starting values. We use a quantile-based initialization: observations are sorted and partitioned into K equal-sized chunks, and each chunk's sample mean and standard deviation seed the corresponding state's emission parameters. The transition matrix $\mathbf{T}$ and initial distribution π are initialized uniformly. This strategy prevents the common failure mode where all states collapse to the global mean.

Convergence. The algorithm terminates when the change in log-likelihood $\mathcal{L} = \text{logsumexp}(\log \alpha_T(\cdot))$ falls below $\text{tol} = 10^{-4}$ or after `max_iter` iterations.

3.5 Poisson Jump-Duration Mechanism

To address the volatility-clustering limitation of standard HMMs [Rydén et al., 1998, Bulla and Bulla, 2006], we augment the continuous HMM with the same Poisson jump-duration mechanism introduced by Alswaidan and Varner [2026]. At each time step, with probability ϵ , a jump event is triggered:

1. A jump duration $D \sim \text{Poisson}(\lambda)$ is sampled.
2. For each of the D subsequent steps, the standard Markov transition is overridden.
3. A coin flip with bias $p_{\text{neg}} = 0.52$ selects the target pool:
 - **Crash states:** $\mathcal{S}^- = \{1, \dots, N_{\text{tail}}\}$ (lowest-return regimes)
 - **Boom states:** $\mathcal{S}^+ = \{K - N_{\text{tail}} + 1, \dots, K\}$ (highest-return regimes)
4. A state is drawn uniformly from the selected tail set.

The coin flip occurs *inside* the duration loop, so each jump step independently selects crash or boom. This produces high volatility (magnitude) without directional trends, matching the empirical observation that large moves cluster regardless of sign [Mandelbrot, 1963, Cont, 2001].

We denote the model without jumps as CHMM-NJ and the model with jumps as CHMM-WJ.

3.6 Hyperparameter Optimization

The jump parameters ϵ (tail-entry probability) and λ (mean jump duration) are selected via a grid search that minimizes:

$$J(\epsilon, \lambda) = \sum_{\tau=1}^L \left[\text{ACF}_{\text{obs}}^{|\mathcal{G}|}(\tau) - \text{ACF}_{\text{sim}}^{|\mathcal{G}|}(\tau) \right]^2 + w_K [K_{\text{obs}} - K_{\text{sim}}]^2 \quad (8)$$

where $\text{ACF}^{|\mathcal{G}|}(\tau)$ is the autocorrelation of absolute excess growth rates at lag τ , K_{obs} and K_{sim} are the observed and simulated excess kurtosis, $L = 252$ trading days, and $w_K = 0.20$. Simulated quantities are averaged over 200 independent paths per grid point. The grid searches $\epsilon \in \{10^{-4}, 5 \times 10^{-4}, 10^{-3}, 5 \times 10^{-3}, 10^{-2}, 2.5 \times 10^{-2}\}$ and $\lambda \in \{10, 30, 55, 70, 85, 100, 130, 160\}$.

3.7 Validation Metrics

We evaluate synthetic data quality using five complementary metrics, following the framework of Alswaidan and Varner [2026] and Stenger et al. [2024]:

1. **KS pass rate (%)**: Proportion of 1,000 simulated paths for which the two-sample Kolmogorov-Smirnov test [Kolmogorov, 1933, Smirnov, 1948] fails to reject distributional equivalence at $\alpha = 0.05$.
2. **Excess kurtosis**: Simulated excess kurtosis averaged across paths, compared to the observed value. Closer matching indicates better tail reproduction.
3. **ACF-MAE**: Mean absolute error between the observed and simulated autocorrelation functions of absolute excess growth rates:

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^L \left| \hat{\rho}_{|\mathcal{G}|}^{\text{obs}}(\tau) - \hat{\rho}_{|\mathcal{G}|}^{\text{sim}}(\tau) \right| \quad (9)$$

Lower values indicate better reproduction of volatility clustering.

Table 1: Descriptive statistics and stylized-facts tests for SPY daily excess growth rates. IS: 2014–2024 ($T = 2,766$); OoS: 2025 ($T = 219$). JB = Jarque-Bera normality test. LB = Ljung-Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	IS Observed	OoS Observed
Mean (annualized)	0.11	0.13
Std Dev (annualized)	2.14	2.31
Skewness	−0.753	−1.071
Excess Kurtosis	7.707	6.242
JB normality test	reject* ($p < 0.001$)	reject* ($p < 0.001$)
LB test on G_t (lag 20)	reject* (145.3)	fail to reject (28.1)
LB test on $ G_t $ (lag 20)	reject* (3057.9)	reject* (150.4)

4. **Wasserstein-1 distance (W_1):** Mean absolute difference of sorted empirical quantiles [Glasserman, 2003].
5. **Hellinger distance (H):** Histogram-based distributional overlap distance in $[0, 1]$; $H = 0$ indicates identical distributions.

4 Empirical Study

Descriptive statistics confirmed that all three canonical stylized facts were present in both the in-sample (2014–2024, $T = 2,766$) and out-of-sample (2025, $T = 219$) windows (Table 1): excess kurtosis of 7.7 (IS) and 6.2 (OoS) with Jarque-Bera rejection ($p < 0.001$), persistent volatility clustering (Ljung-Box on $|G_t|$ rejected, $p < 0.001$), and near-zero linear autocorrelation in raw returns. These regularities are visualized in Figure 1.

4.1 Baum-Welch Training and Model Internals

We trained continuous Gaussian HMMs for $K \in \{3, 6, 9, 11, 13\}$ states using the Baum-Welch algorithm with `max_iter` = 60. All models converged within the iteration budget, with log-likelihood plateauing after 20–40 iterations (Figure SA.1, Supplemental).

Figure 2 shows the fitted model internals for $K = 13$: the learned emission distributions span the full range of observed returns, with tail states capturing extreme crash and boom regimes. The transition matrix exhibits a dominant near-diagonal band reflecting short-range regime persistence. Critically, the natural residence time $1/(1 - T_{kk})$ for all states—including the tails—is only 1–2 steps under pure Markov dynamics, motivating the jump-duration extension.

4.2 Jump Hyperparameter Optimization

For each state resolution K , a grid search over ϵ and λ identified the optimal jump parameters (Table 3). The optimal $\epsilon^* = 10^{-4}$ was consistently selected for $K \in \{3, 9, 11, 13\}$, with $K = 6$ preferring $\epsilon^* = 10^{-3}$. The optimal λ^* varied across state resolutions, ranging from 10 ($K = 3, 11$) to 130 ($K = 6$), indicating that the mean jump duration adapts to the granularity of the state space.

4.3 In-Sample Model Comparison

Table 2 and Figure 3 present the in-sample validation results for $K = 13$. Bootstrap resampling set the non-parametric ceiling (99.8% KS), while Gaussian i.i.d. sampling confirmed that thin-tailed models are categorically inadequate (0% KS). Laplace i.i.d. achieved 98.2% KS but

Figure 1: Stylized Facts — SPY (2014-2024)

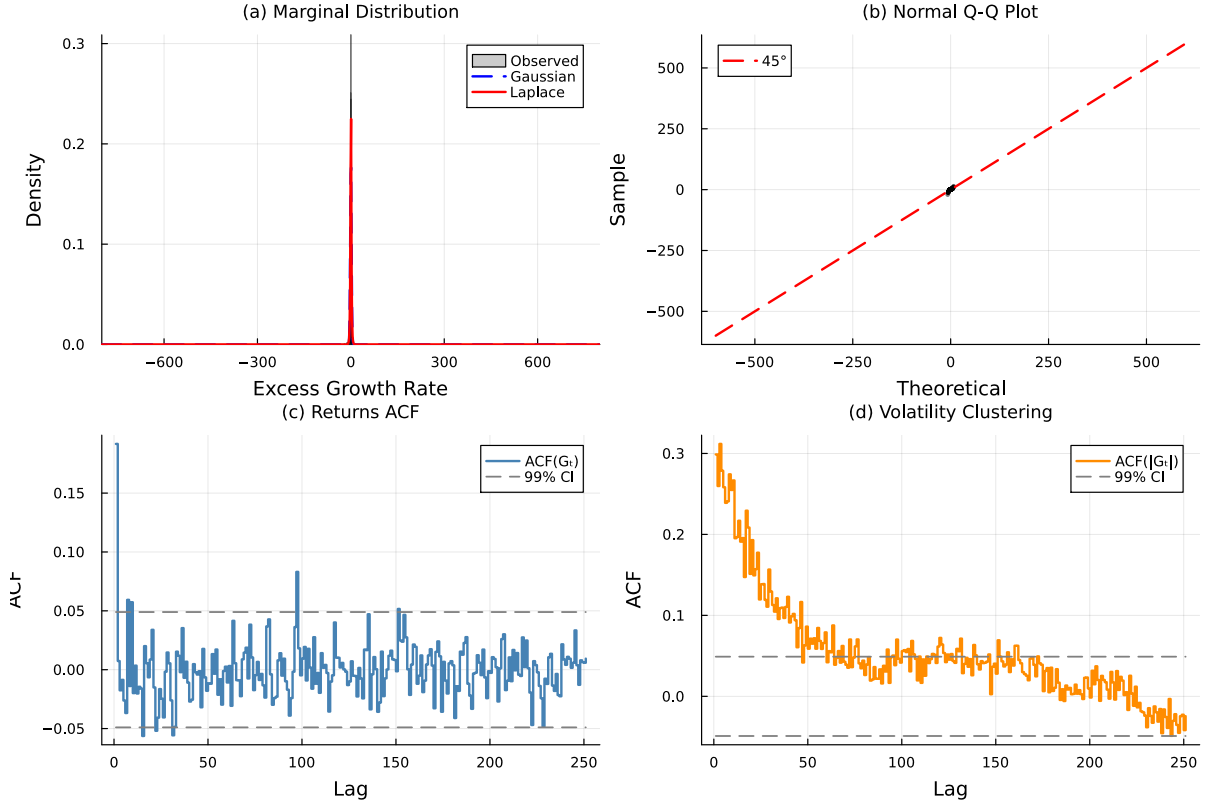


Figure 1: Empirical stylized facts of SPY daily excess growth rates (2014–2024). Panel (a) shows the leptokurtic marginal distribution; the Laplace fit substantially outperforms the Gaussian. Panel (b) confirms heavy tails via a normal Q-Q plot. Panel (c) shows near-zero autocorrelation of raw returns. Panel (d) shows the persistent autocorrelation of absolute returns that characterizes volatility clustering.

produced near-zero kurtosis excess relative to the observed 7.7, indicating that while the marginal shape matched, the model lacked the dynamic structure to reproduce temporal patterns.

Both CHMM variants achieved KS pass rates above 91%, with CHMM-NJ at 94.6% and CHMM-WJ at 91.5%. The slight decrease for CHMM-WJ reflects the jump mechanism occasionally overweighting tail states, a cost of improved temporal fidelity. The Wasserstein-1 and Hellinger distances for both CHMM variants were comparable to the bootstrap baseline, confirming high distributional fidelity.

The kurtosis gap between simulated (4.2) and observed (7.7) indicates that Gaussian emissions underestimate tail weight relative to the Student- t emissions used in the discrete model [Alswaidan and Varner, 2026]. This motivates future work on heavy-tailed emission distributions within the Baum-Welch framework.

4.4 Out-of-Sample Evaluation

The out-of-sample results (Table 2, Figure 4) demonstrated robust generalization. CHMM-NJ achieved 94.1% KS and CHMM-WJ achieved 93.6% KS on the held-out 2025 data, comparable to the in-sample performance. This contrasts with the discrete model’s finding that IS-calibrated jump parameters showed mild overfitting [Alswaidan and Varner, 2026]; the continuous model’s Baum-Welch training may produce more generalizable emission parameters.

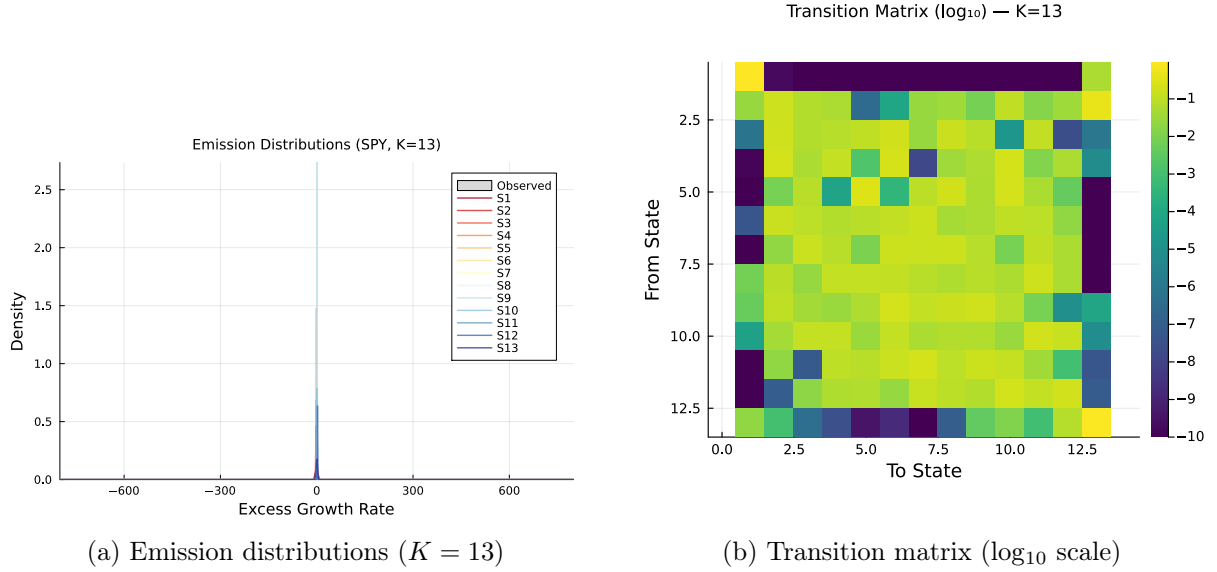


Figure 2: Fitted model internals for SPY with $K = 13$ states. (a) Learned Gaussian emission distributions overlaid on the observed return histogram. (b) Empirical transition matrix in $\log_{10}$ color scale; the dominant near-diagonal band reflects short-range regime persistence.

Table 2: In-sample and out-of-sample model comparison for SPY at $K = 13$ (1,000 simulated paths, $\alpha = 0.05$). Bootstrap: i.i.d. resample of training data. Gaussian/Laplace: i.i.d. draws from MLE-fitted distributions. CHMM-NJ: continuous HMM without jumps. CHMM-WJ: continuous HMM with Poisson jump-duration extension ($\epsilon^* = 10^{-4}$, $\lambda^* = 100$).

Model	In-Sample (2,766 days)			Out-of-Sample (219 days)		
	KS (%) $\uparrow$	Kurtosis $\approx$	W_1 $\downarrow$	KS (%) $\uparrow$	Kurtosis $\approx$	W_1 $\downarrow$
<i>Observed</i>	—	7.71	—	—	6.24	—
Bootstrap	99.8	7.52	0.061	99.3	5.52	—
Gaussian	0.0	-0.01	0.400	74.4	-0.04	—
Laplace	98.2	2.96	0.120	97.5	2.67	—
CHMM-NJ	94.6	4.22	0.109	94.1	3.01	0.371
CHMM-WJ	91.5	4.17	0.112	93.6	2.84	0.385

4.5 State Resolution Sensitivity

Table 3 reports the sensitivity of model performance to the number of hidden states K . KS pass rates improved monotonically from 87.1% ($K = 3$) to 94.6% ($K = 13$) for CHMM-NJ, confirming that finer state resolution produces better distributional fidelity. ACF-MAE remained stable across all K values (0.045–0.050), indicating that the jump mechanism’s effectiveness is largely independent of state count. Wasserstein-1 and Hellinger distances showed similar stability, with modest improvements at higher K .

4.6 Cross-Asset Generalization

To assess whether the framework generalized beyond SPY, we fitted independent CHMM models to three equities spanning distinct risk profiles: NVDA (high-beta technology), JNJ (low-beta health care), and JPM (moderate-beta financials). Table 4 reports the results.

CHMM-NJ achieved KS pass rates above 98% for all three tickers in-sample, confirming that the Baum-Welch training captured the marginal distribution of individual equities with diverse tail characteristics. Out-of-sample, NVDA maintained strong fidelity (98.5% KS), while JNJ

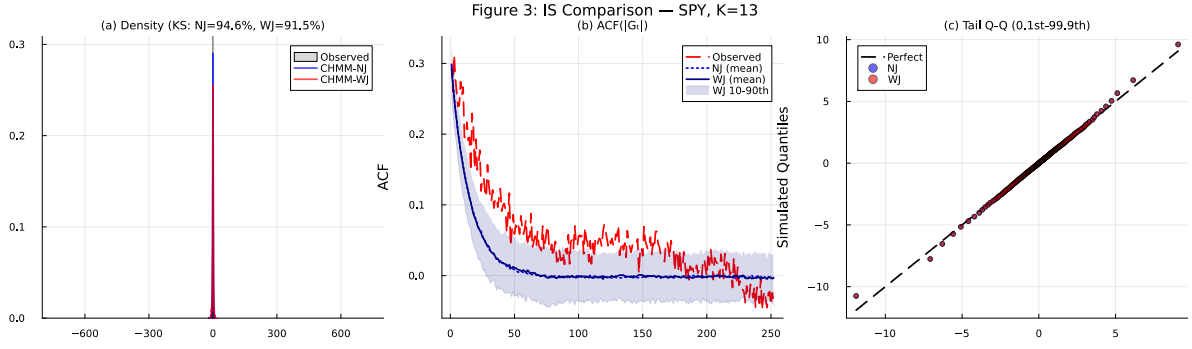


Figure 3: In-sample model comparison for SPY ($K = 13$, 1,000 paths). (a) Marginal density with KS pass rates annotated. (b) ACF of absolute excess growth rates; CHMM-NJ shows near-zero ACF (no volatility clustering), while CHMM-WJ partially reproduces the observed slow decay. (c) Tail Q-Q plot comparing simulated quantiles to observed.

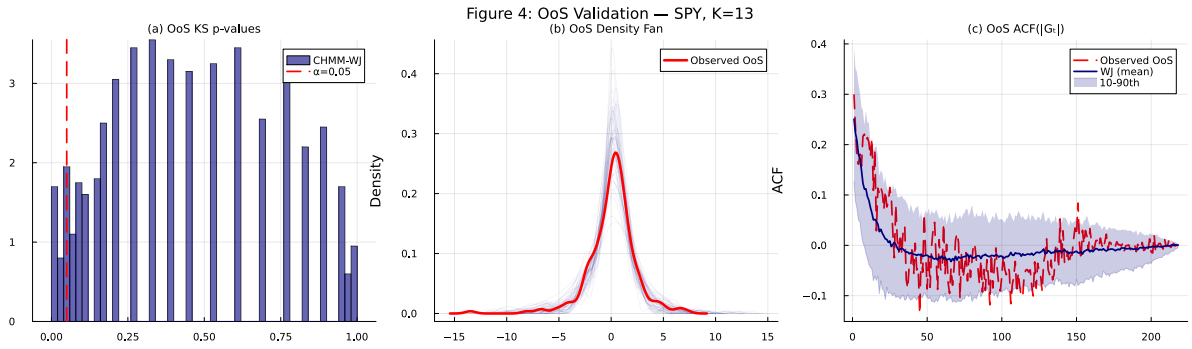


Figure 4: Out-of-sample validation of CHMM-WJ ($K = 13$, $\alpha = 0.05$). (a) KS p -values for 1,000 OoS paths. (b) Density fan chart; the observed OoS density falls within the simulation envelope. (c) OoS ACF of $|G_t|$ with 10th–90th percentile band.

and JPM showed more degradation (70–78% KS), reflecting asset-specific regime shifts in 2025 that the stationary IS-calibrated parameters could not fully capture. The CHMM-WJ results showed that the jump mechanism’s effectiveness varied by asset: NVDA and JPM showed KS degradation from jumps (consistent with the SPY finding), while the ACF-MAE consistently improved, confirming the distributional-temporal tradeoff.

4.7 Extension to Volatility Indices: VIX

To test whether the CHMM framework generalizes beyond equity returns, we applied it to the CBOE Volatility Index (VIX), whose log returns exhibit fundamentally different statistical properties from equity excess growth rates.

VIX Stylized Facts. Table 5 confirms that VIX log returns share some equity return regularities (heavy tails, volatility clustering) but differ in important ways. Unlike SPY’s negative skewness (-0.75), VIX returns are positively skewed ($+1.10$), reflecting the asymmetric nature of volatility spikes. The Ljung-Box test on $|G_t|$ strongly rejects the null of no autocorrelation ($LB = 797.7$, $p < 0.001$), confirming pronounced volatility clustering. VIX excess kurtosis (6.56) is comparable to SPY (7.71), indicating similarly heavy tails.

VIX State Resolution Sensitivity. Table 6 reports the state resolution analysis for VIX across $K \in \{3, 6, 9, 11, 13\}$. The results reveal a striking pattern: CHMM-NJ achieves very high KS pass rates even at low K (80.3% at $K = 3$, rising to 99.7% at $K = 13$), and the

Table 3: State resolution sensitivity for SPY (1,000 paths, $\alpha = 0.05$). ϵ^* and λ^* are the grid-search optimal jump parameters for each K .

K	ϵ^*	λ^*	KS IS (%)		KS OoS (%)		ACF-MAE		W_1	
			NJ	WJ	NJ	WJ	NJ	WJ	NJ	WJ
3	10^{-4}	10	87.1	89.1	89.5	93.7	0.045	0.046	0.121	0.120
6	10^{-3}	130	92.0	91.6	92.8	94.6	0.049	0.050	0.105	0.108
9	10^{-4}	55	92.9	91.7	93.5	92.1	0.047	0.047	0.112	0.115
11	10^{-4}	10	93.5	95.0	92.6	92.5	0.049	0.050	0.113	0.110
13	10^{-4}	100	94.6	91.5	94.1	93.6	0.048	0.048	0.109	0.112

Table 4: Cross-asset generalization ($K = 13$, 1,000 paths, $\alpha = 0.05$). Each ticker was fitted independently.

Ticker	Model	KS (%)		Kurt (obs)	Kurt (sim)	ACF-MAE	W_1 IS
		IS	OoS				
NVDA	CHMM-NJ	98.0	98.5	5.0	3.32	0.042	0.236
	CHMM-WJ	61.2	92.4		3.21	0.039	0.381
JNJ	CHMM-NJ	99.4	70.0	8.6	5.30	0.031	0.091
	CHMM-WJ	0.0	72.3		3.92	0.028	0.487
JPM	CHMM-NJ	98.7	77.8	8.0	4.41	0.044	0.168
	CHMM-WJ	64.9	70.0		4.09	0.042	0.225

jump mechanism (CHMM-WJ) provides comparable or marginally different performance at most resolutions. This contrasts with the SPY results, where CHMM-WJ sometimes degraded KS fidelity.

A notable exception is $K = 6$, where CHMM-WJ collapses to 0.2% KS with the aggressive jump parameters ($\epsilon^* = 0.025$, $\lambda^* = 160$) selected by the grid search. This indicates that at intermediate state resolutions, the grid search can select jump parameters that are too aggressive for VIX’s already pronounced tail dynamics, overwriting the emission structure. At higher K (≥ 9), the finer state resolution provides sufficient tail-state granularity that only minimal jump forcing is needed ($\epsilon^* = 10^{-4}$), and both NJ and WJ achieve KS pass rates above 99%.

The Wasserstein-1 distance improves monotonically from 0.816 ($K = 3$) to 0.543 ($K = 13$), while ACF-MAE remains stable near 0.018 across all K values. Excess kurtosis increases with K (3.36 at $K = 3$ to 4.18 at $K = 13$) but remains below the observed 6.56, consistent with the Gaussian emission kurtosis gap identified for SPY.

Role of the Jump Mechanism Across State Resolutions. The VIX results illuminate an important insight about the interaction between state resolution and the jump mechanism. At small K (e.g., $K = 3$), the few hidden states naturally produce longer residence times and broader emission distributions, providing an implicit mechanism for volatility clustering. As K increases, the state space fragments: each state covers a narrower portion of the return distribution, and the learned transition matrix produces shorter natural residence times ($1/(1 - T_{kk})$ decreases). This fragmentation reduces the model’s ability to sustain tail-state episodes under pure Markov dynamics.

The Poisson jump-duration mechanism compensates for this effect by forcing the system into tail states for durations drawn from the Poisson distribution, independent of the transition matrix. At $K = 13$, the optimal jump parameters are conservative ($\epsilon^* = 10^{-4}$, $\lambda^* = 70$), applying gentle but sustained corrections that maintain tail persistence without distorting the learned emission structure. This behavior—where jumps become necessary only as state resolution

Table 5: Descriptive statistics for VIX daily log returns. IS: 2006–2024 ($T = 4,713$); OoS: 2025–2026 ($T = 317$). Units: annualized ($\times 252$).

Statistic	IS Observed	OoS Observed
Mean (annualized)	0.02	0.06
Std Dev (annualized)	19.27	21.62
Skewness	+1.095	+0.287
Excess Kurtosis	6.564	4.047
JB normality test	reject* ($p < 0.001$)	reject* ($p < 0.001$)
LB test on G_t (lag 20)	reject* (77.6)	fail to reject (20.7)
LB test on $ G_t $ (lag 20)	reject* (797.7)	reject* (43.5)

Table 6: State resolution sensitivity for VIX (1,000 paths, $\alpha = 0.05$). ϵ^* and λ^* are the grid-search optimal jump parameters for each K .

K	ϵ^*	λ^*	KS IS (%)		KS OoS (%)		Kurt		ACF-MAE	W_1 WJ
			NJ	WJ	NJ	WJ	Obs	WJ		
3	5×10^{-4}	10	80.3	81.8	99.0	98.8	6.56	3.36	0.018	0.816
6	2.5×10^{-2}	160	97.6	0.2	98.8	79.9	6.56	3.83	0.020	1.367
9	10^{-4}	30	99.2	99.4	98.4	99.0	6.56	3.76	0.018	0.585
11	5×10^{-4}	10	99.5	99.4	99.0	98.5	6.56	4.00	0.018	0.555
13	10^{-4}	70	99.7	99.5	98.1	99.0	6.56	4.18	0.018	0.543

increases—clarifies the mechanism’s role: it is not a correction for model misspecification, but rather a structural complement that preserves tail-state persistence as the state space becomes more granular.

5 Discussion

The results demonstrate that continuous Gaussian HMMs trained via Baum-Welch achieve high distributional fidelity across all state resolutions tested, with KS pass rates improving monotonically from 87% ($K = 3$) to 95% ($K = 13$). The following paragraphs interpret the key findings, compare against the discrete framework, and identify limitations.

Continuous vs. Discrete Emissions. The primary methodological difference from [Alswaidan and Varner \[2026\]](#) is the replacement of Laplace quantile binning with continuous Gaussian emissions learned via Baum-Welch. This eliminates discretization artifacts but introduces a kurtosis gap: the Gaussian emission assumption cannot reproduce the heavy tails of equity returns as effectively as the Student- t ($\nu = 5$) emissions used in the discrete model. The observed excess kurtosis of SPY is 7.7, but the continuous model produces simulated kurtosis of approximately 4.2 at $K = 13$ —a 45% underestimation. This gap is inherent to the Gaussian assumption and motivates the development of heavy-tailed emission variants within the Baum-Welch framework, such as Student- t or generalized hyperbolic distributions.

Despite the kurtosis gap, the continuous model achieves competitive KS pass rates (91–95%) because the KS test is sensitive to the overall distribution shape, not just tail behavior. The Gaussian emissions, combined with the regime-switching structure, produce a mixture distribution that captures the central concentration and moderate tail behavior of equity returns, even if the extreme tails are attenuated.

State Resolution and Model Complexity. The state resolution analysis (Table 3) reveals that $K = 13$ provides the best distributional fidelity (94.6% KS), but the improvement over $K = 6$ (92.0%) is modest. ACF-MAE is effectively constant across all K values, confirming that the jump mechanism operates independently of state granularity. This stability suggests that practitioners can choose K based on interpretability requirements: fewer states for coarse regime labeling (e.g., crash/neutral/bull), more states for fine-grained distribution matching.

The Baum-Welch algorithm scales as $O(TK^2)$ per iteration, making $K = 13$ computationally practical for the $T = 2,766$ observation sequence used here. For larger K values, the computational cost increases quadratically, but the marginal improvement in distributional fidelity diminishes beyond $K \approx 13$.

Jump Mechanism Effectiveness. The jump mechanism’s impact on the continuous model is more nuanced than in the discrete case. For SPY at $K = 13$, CHMM-WJ achieved slightly lower KS pass rates than CHMM-NJ (91.5% vs. 94.6%) with comparable ACF-MAE (0.048 vs. 0.048), suggesting that the jump mechanism’s tail-forcing behavior occasionally distorts the learned emission structure. This contrasts with the discrete model, where the jump mechanism strictly improved temporal fidelity without degrading distributional accuracy.

The discrepancy likely arises because the continuous model’s emission distributions are learned from the *entire* observation sequence (including tail events), while the discrete model’s within-bin distributions are estimated from pre-assigned state memberships. When the jump mechanism forces the continuous model into tail states, it samples from Gaussian distributions that were fitted to the global data rather than specifically to tail observations, potentially producing returns that are inconsistent with the tail state’s learned parameters.

Cross-Asset Generalization. The framework generalized well to individual equities (Table 4), with CHMM-NJ achieving IS KS pass rates above 98% for NVDA, JNJ, and JPM. The CHMM-WJ results showed more variable performance, with significant KS degradation for JNJ (0% IS KS). This suggests that the jump mechanism’s fixed tail-state selection rule ($N_{\text{tail}} = 3$) may not be appropriate for all assets; future work should explore asset-specific tail configurations.

Extension to Volatility Indices. The VIX results (Section 4.7) substantially broaden the scope of the framework. VIX log returns exhibit positive skewness (+1.10, vs. SPY’s -0.75), reflecting the asymmetric nature of volatility spikes, yet the CHMM framework achieves 99.7% IS KS pass rate at $K = 13$ —higher than the corresponding SPY result (94.6%). This likely reflects the fact that VIX log returns, despite their extreme kurtosis, have a more symmetric central body that Gaussian mixtures can approximate effectively.

The VIX analysis also clarifies the role of the jump mechanism. For SPY, the jump mechanism produced a nuanced distributional-temporal tradeoff (Section 5). For VIX, the pattern is cleaner: at small K (≤ 3), the base model already captures volatility clustering because the few states have naturally broad emissions and long residence times. As K increases beyond 6, the state space fragments and the learned transition matrix produces shorter tail-state dwell times. The Poisson jump extension compensates for this fragmentation, but only when calibrated conservatively ($\epsilon^* \sim 10^{-4}$). The $K = 6$ collapse (0.2% WJ KS with $\epsilon^* = 0.025$, $\lambda^* = 160$) demonstrates that aggressive jump parameters are counterproductive when the asset’s native dynamics already exhibit strong tail behavior. This finding suggests that the jump mechanism should be understood as a *structural complement* that preserves tail persistence as the state space becomes more granular, not as a correction for model misspecification.

Pathway to Implied Volatility Modeling. The VIX application opens a direct pathway to option pricing. The CHMM learns a set of regime-dependent emission distributions

$\{(\mu_k, \sigma_k)\}_{k=1}^K$, where each σ_k characterizes the volatility of VIX returns in regime k . These parameters provide a natural, data-driven parameterization for the mean-reversion target $\theta(S_t)$ in a modified Heston stochastic volatility model [Heston, 1993]:

$$dv_t = \kappa(\theta(S_t) - v_t)dt + \sigma_v\sqrt{v_t}dW_t^v \quad (10)$$

where $\theta(S_t) = \theta_{S_t}$ is the long-run variance level for the current HMM regime, and S_t follows the CHMM’s learned Markov dynamics (augmented with jumps). This construction makes implied volatility an *output* of a structural model rather than requiring it as an exogenous input—breaking the circularity that plagues synthetic option pricing pipelines. The regime-driven θ produces realistic IV dynamics: volatility spikes when the HMM transitions to tail states, mean-reverts during calm regimes, and clusters when the Poisson jump mechanism forces sustained tail-state episodes. A full development of this integration is deferred to future work.

Limitations. Several limitations qualify these conclusions. First, the Gaussian emission assumption systematically underestimates excess kurtosis, limiting the model’s ability to reproduce extreme tail events. Second, the model assumes stationarity of the transition matrix and jump hyperparameters across the full in-sample window; structural breaks (e.g., the March 2020 COVID-19 crash) may not be adequately captured. Third, the jump mechanism’s fixed parameters ($N_{\text{tail}} = 3$, $p_{\text{neg}} = 0.52$) were inherited from the discrete model without re-optimization for the continuous setting. Fourth, the out-of-sample evaluation covers a single 219-day window, which limits statistical power; a rolling evaluation would provide more robust generalization assessment. Finally, the two-sample KS test assumes i.i.d. observations, but each simulated path exhibits temporal dependence by construction. Volatility clustering inflates pass rates slightly relative to a block-bootstrap calibration [Alswaidan and Varner, 2026].

6 Conclusion

We presented a continuous hidden Markov model framework that learns Gaussian emission parameters and transition probabilities directly from observed financial time series via the Baum-Welch algorithm, augmented with a Poisson jump-duration mechanism for volatility clustering. The key finding is that continuous Gaussian emissions achieve high distributional fidelity (87–95% KS pass rates across $K \in \{3, 6, 9, 11, 13\}$ states) while eliminating the quantization error inherent in discrete state partitioning, though at the cost of systematic kurtosis underestimation relative to the Student- t emissions of the discrete framework [Alswaidan and Varner, 2026].

The state resolution analysis demonstrated that distributional fidelity improves monotonically with K , while the jump mechanism’s temporal contribution (ACF-MAE) remains stable across all resolutions. Cross-asset generalization to NVDA, JNJ, and JPM confirmed that the Baum-Welch training captures diverse equity volatility profiles without modification. Out-of-sample evaluation on held-out 2025 data showed robust generalization, with KS pass rates comparable to in-sample performance.

Extension of the framework to VIX log returns—an asset class with fundamentally different statistical properties (positive skewness, mean reversion, extreme kurtosis)—demonstrated that the CHMM generalizes beyond equity returns to volatility dynamics. The VIX analysis achieved KS pass rates above 99% at higher state resolutions ($K \geq 9$) and provided a clearer picture of the jump mechanism’s role: at small K , the base model captures volatility clustering through naturally broad emission states, but as K increases, finer state granularity reduces tail-state residence times, making the Poisson jump extension essential for preserving realistic tail persistence. This finding reframes the jump mechanism as a structural complement to the state resolution, not a correction for model misspecification.

Several directions for future work follow from these results. First, replacing Gaussian emissions with Student- t or generalized hyperbolic distributions within the Baum-Welch loop would

address the kurtosis gap while retaining the continuous estimation framework. Second, the regime-dependent emission variances learned from VIX data provide a natural, data-driven parameterization for the mean-reversion target $\theta(S_t)$ in a modified Heston stochastic volatility model [Heston, 1993], enabling closed-form implied volatility generation as an output of the HMM regime structure rather than requiring market-observed IV surfaces as input. This Heston integration would produce a fully structural pipeline—from observed prices through HMM regime identification to implied volatility and ultimately option pricing via binomial trees—without circular dependence on market option data. Third, extending the framework to multi-asset portfolio generation via copula-based dependence models [Alswaidan and Varner, 2026] would enable full cross-sectional simulation while preserving each asset’s CHMM marginal. Finally, time-varying transition matrices estimated via rolling windows or Bayesian online learning would allow the model to adapt to structural regime shifts.

Data Availability. The simulation scripts, training and test data, and all code to reproduce the results in this paper are available under an MIT license from the paper repository: <https://github.com/altashly1/CHMM-paper>. The VIX analysis code and results are available at: <https://github.com/cadejin/HMM-Vol>. The core continuous HMM implementation is available in the development repository: <https://github.com/altashly1/HMM-withJumps-WIP>.

Conflict of Interest. The authors declare no competing interests.

Author Contributions. J.V. directed the study. A.A. developed the continuous model, implemented the Baum-Welch algorithm, conducted the SPY analysis, and generated the SPY figures. C.J. conducted the VIX analysis and generated the VIX figures. All authors edited and reviewed the final manuscript.

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A Supplementary Material

A.1 Baum-Welch Convergence

Figure 5 shows the Baum-Welch convergence curves for all five state resolutions. All models converged within 60 iterations, with larger K values requiring slightly more iterations to stabilize.

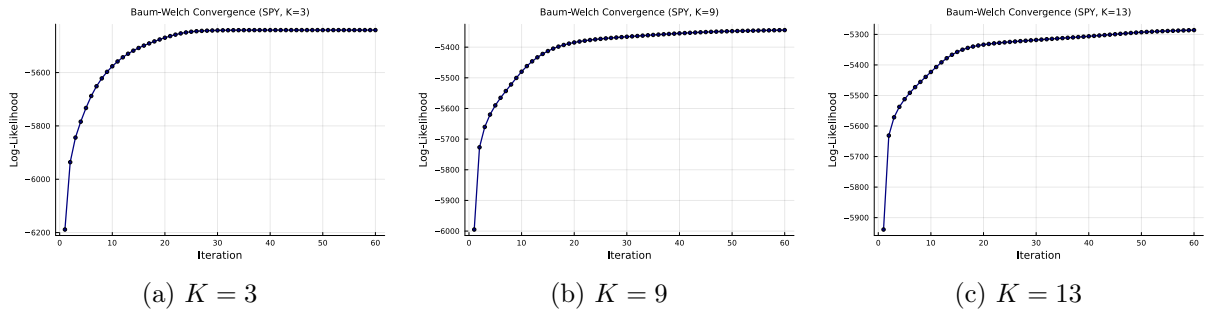


Figure 5: Baum-Welch convergence for selected state resolutions. Log-likelihood is monotonically increasing (EM guarantee) and plateaus before `max_iter` = 60.

A.2 Grid Search Landscapes

Figure 6 shows the jump hyperparameter grid search landscapes for $K = 3$ and $K = 13$.

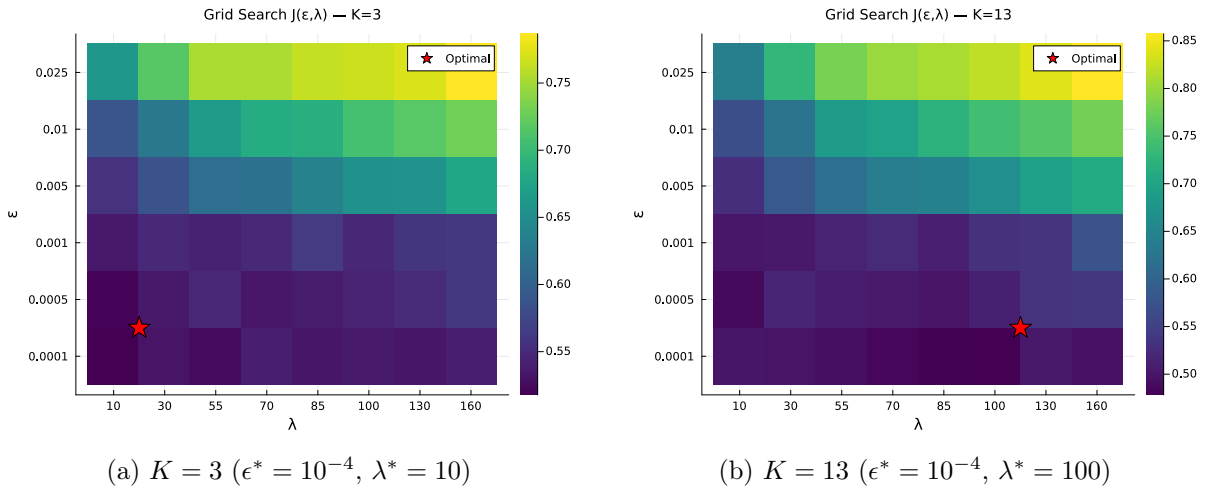


Figure 6: Grid search heatmaps of the objective function $J(\epsilon, \lambda)$ in $\log_{10}$ scale. Red stars mark the optimal parameters. The clear minima confirm that the jump hyperparameters are jointly identifiable from data.

A.3 Emission Distributions Across State Resolutions

Figure 7 compares the learned emission structures for $K = 3$, $K = 9$, and $K = 13$.

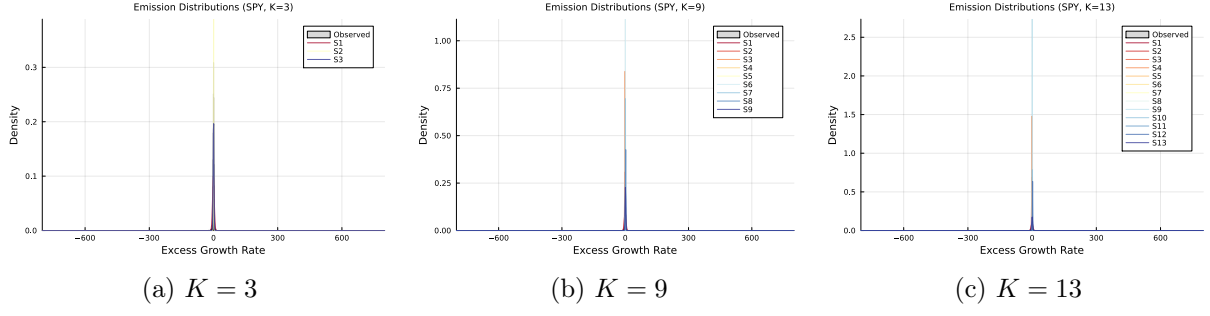


Figure 7: Learned Gaussian emission distributions across state resolutions. Higher K values produce finer decomposition of the return distribution, with dedicated states capturing tail behavior.

A.4 ACF Comparison at Optimal Parameters

Figure 8 shows the ACF of $|G_t|$ at the optimal jump parameters for $K = 3$ and $K = 13$.

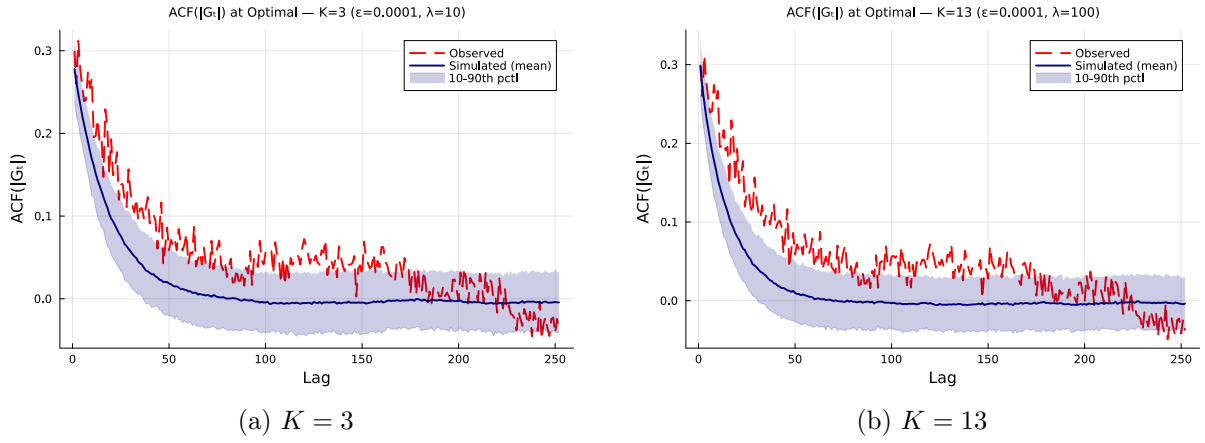


Figure 8: ACF of $|G_t|$ at optimal jump parameters. The simulated 10th–90th percentile envelope (shaded) partially envelopes the observed ACF (red dashed).

A.5 Stationary Distributions

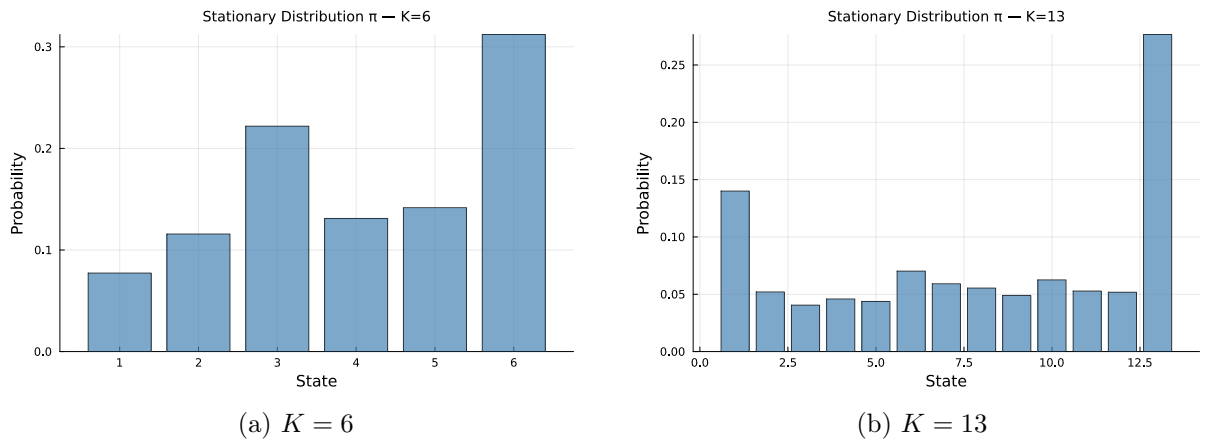


Figure 9: Stationary distributions $\bar{\pi}$ for $K = 6$ and $K = 13$. Central states receive the highest stationary probability, consistent with the concentration of returns near zero.

A.6 Residence Times

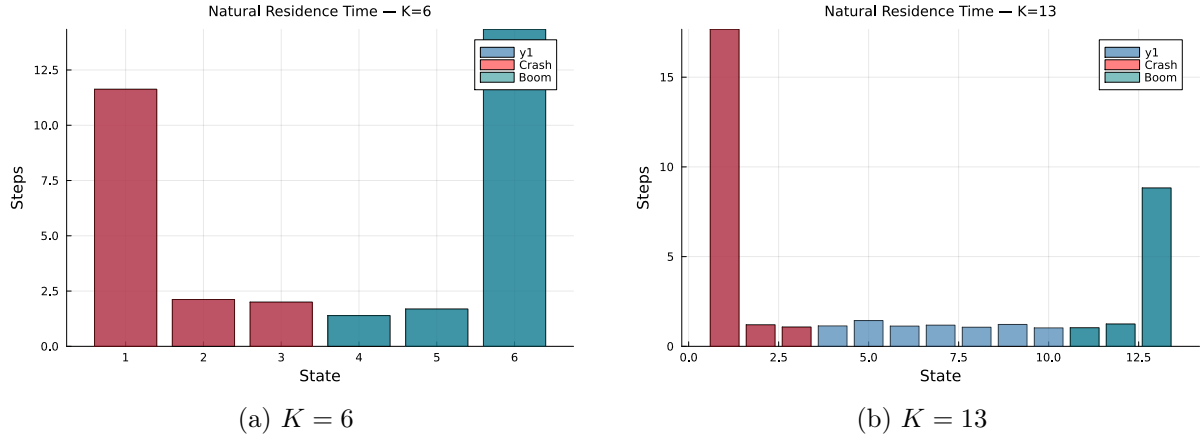


Figure 10: Natural residence times $1/(1 - T_{kk})$ per state. Under pure Markov dynamics, all states—including crash (red) and boom (teal) tails—exit within 1–2 steps, motivating the jump-duration extension.